



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	4YRS B.S	IIT Kharagpur	8.43 / 10
2023	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	91.2%
2021	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	97.6%

WORK EXPERIENCES

Geological Field Trip | Academic

- Applied geophysical signal processing, spectral analysis, frequency-domain methods, and digital filtering to interpret geological structures.
- Used Python, NumPy, and Matplotlib for geophysical data processing, spectral analysis, visualization, and quantitative interpretation.
- Analyzed frequency spectra and signal characteristics to identify geological features and distinguish meaningful signals from noise.

Seismic Field Trip | Academic

- Acquired, processed, and interpreted seismic field data to identify subsurface geological structures and characterize seismic responses.
- Applied signal processing, statistical analysis, and visualization techniques to evaluate seismic measurements and subsurface responses.
- Interpreted seismic signatures and structural features to analyze subsurface geometry and support geological mapping.

Electrical Field Trip | Academic

- Conducted electrical resistivity surveys using **multiple electrode arrays** and **ERT** to investigate subsurface geological lithologies
- Applied **ERT inversion, statistical analysis, and visualization** to characterize subsurface resistivity and structural features.
- Analyzed electrical resistivity variations to identify geological boundaries, subsurface structures, and anomalous zones.

Radiometric Field Trip | Academic

- Conducted radiometric surveys and quantitative analysis to characterize geological variations and identify radiometric anomalies.
- Processed and statistically analyzed radiometric measurements to support geological interpretation and subsurface mapping.
- Interpreted gamma-ray signatures and spatial variations to support lithological characterization and geological mapping.

PROJECTS

3-Phase Black-Oil Simulator for Heterogeneous Unconventional Reservoirs | BTP Project | Prof. Pares Nath Singha Roy

- Built a 3-phase black oil reservoir simulator in Python solving fully implicit PDEs with **Newton-Raphson** iteration and **Peaceman** well model.
- Implemented **fractal** based porosity distribution to model unconventional reservoir heterogeneity and improve geological realism.
- Reduced simulation runtime by 3x by using **Random Forest** to provide close initial estimates of FVF, viscosity, and relative permeability
- Analyzed impact of fractal dimension on pressure depletion and cumulative oil/gas recovery under water-flood scenarios.

INTERNSHIPS

Data Science Intern | Polestar Analytics

May 2026 - Jul 2026

Project: Agentic Stock Manager | Human-in-the-loop system for automated supply-chain replenishment

- Built a **multi-agent system** to detect stock threshold breaches and generate **purchase orders** from live inventory, warehouse, and product data.
- Integrated real-time APIs, multi-LLM routing, purchase order approvals, guardrails, and audit logs into a **6-agent LangGraph workflow**.
- Consolidated per-SKU orders into **one purchase order per vendor** using vendor based batching, supporting auto issue, draft, batch, and suppress

Project: Promotional Uplift Modeling | Time-aware forecasting and causal promotion targeting

- Built a time aware demand forecasting pipeline integrating multi-source retail data with lag, rolling, trend, and **Fourier features**.
- Developed and benchmarked **CatBoost T-, S-, and X-learners** alongside hierarchical Bayesian uplift models using **Qini, AUUC, uplift@K**
- Converted uplift estimates into a capacity constrained **promotion policy**, prioritizing product-weeks by predicted incremental demand

Analyst Programmer Intern | Xceedance Consulting Pvt. Ltd.

May 2025 - Sept 2025

- Architected an end to end insurance document pipeline using **Tesseract OCR** for text extraction and **BERT** based NER for entity extraction
- Fine tuned **BERT** to extract insurance entities from OCR text, achieving **94.2%** accuracy and **0.91** weighted **F1**, reducing costly LLM API calls.
- Implemented **RapidFuzz** based entity **normalization** to map noisy NER outputs to standardized reference records, reducing manual review
- Built a **Streamlit** interface for NER and normalization, integrating **MLflow** tracking and automated **Databricks** ML pipeline execution.

Undergraduate Student Researcher | Prof. Pawan Goyal | CNeRG Lab, Department of Computer Science and Engineering, IIT Kharagpur

- Scraped and curated Sanskrit text from Rigveda and Atharvaveda using **BeautifulSoup**, aligning svara verses with audio for **ASR** dataset
- Fine-tuned **Wav2Vec2** and **IndicWhisper** by expanding **vocabularies** to accommodate **Vedic svaras** for Sanskrit speech recognition.
- Evaluated speech-to-text transcription using **Word Error Rate (WER)** and **Character Error Rate (CER)**, achieving **17.5%** WER and **6.2%** CER.

COURSEWORK INFORMATION

Computer Science: Information Retrieval, Machine Learning, Algorithms (Theory & Laboratory) , Programming & Data Structures (Theory & Laboratory) , Operating Systems, Database Management Systems, Deep Learning, Computer Networks, Natural Language Processing**Maths:** Linear Algebra & Numerical Analysis, Advanced Calculus, Probability & Statistics, Partial Differential Equations, Transform Calculus**Core:** Methods of Prospecting, Pattern Recognition, Borehole Geophysics, Geophysical Signal Processing, Geophysical Inverse & Field Theory

EXTRA CURRICULAR ACTIVITIES

- Promoted to Head of Design at Click KGP, IIT Kharagpurs Photography Club, and later served as Advisor.
- Selected for the Inter IIT Badminton Contingent; captained Nehru Halls badminton team in General Championship Sports.
- Achieved peak ratings of 1500+ in Chess.com Rapid, 1600+ in Blitz, and 1600+ in Matiks Sprint Duels.